Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Linux Masterclass

Course Code: C579

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 October 2026	Initial release of the Linux Masterclass lesson plan — 2-day schedule (9:30am–5:30pm) covering the four topics and 17 hands-on labs on the KillerCoda Ubuntu playground.	Dr Alfred Ang

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Course Information

Course Title	Linux Masterclass
Course Code	C579
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-minute lunch)
Mode	Instructor-led, hands-on Linux configuration and shell-scripting labs on the KillerCoda Ubuntu playground (browser-based)
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyze Linux components and coordinate on the Linux release.
- LO2: Configure Linux and select shell scripting for integrating software products.
- LO3: Execute tests on Linux networking and diagnose networking issues.
- LO4: Identify, modify and improve Linux security configuration.

Course Schedule

Day 1 — Linux Fundamental & Shell Scripting

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, course outline, ground rules and the lab platform (KillerCoda Ubuntu playground) (Slides 1–10)
10:00–10:45	45 min	**Topic 1 — Linux Fundamental: what is Linux, kernel and distributions, ways to run Linux, the file system hierarchy, users and permissions (concepts + demo) (Slides 11–26)**
10:45–12:45	120 min	Hands-on: Lab 1: Getting Started with the KillerCoda Ubuntu Playground; Lab 2: Exploring the Linux File System; Lab 3: Working with Files, Directories and Links; Lab 4: Users, Ownership and Permissions (Slides 27–39)
12:45–13:15	30 min	Lunch break
13:15–15:00	105 min	Topic 2 — Linux Shell Script: shell environment, redirection and pipes, grep and regular expressions, first scripts

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		(concepts + demo). Hands-on: Lab 5: Customising the Shell Environment; Lab 6: Redirection, Pipes and Text Processing; Lab 7: Your First Shell Script; Lab 8: User Input, Arguments and Arithmetic (Slides 40–74)
15:00–17:15	135 min	Bash scripting constructs: conditionals, loops, functions, debugging, automation (concepts + demo). Hands-on: Lab 9: Conditionals — if, test and case; Lab 10: Loops — for, while and until; Lab 11: Functions, Exit Status and Debugging; Lab 12: Automation Script — Backup with tar (Slides 75–87)
17:15–17:30	15 min	Day 1 recap and Q&A

9:30am – 5:30pm · 450 minutes (7.5 instructional hours), with a 30-minute lunch.

Day 2 — Networking Configuration & Security

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and Q&A
9:45–11:25	100 min	**Topic 3 — Linux Networking Configuration: processes and cron, package management, service deployment, network configuration tests and diagnosis (concepts + demo) (Slides 88–101)**
11:25–13:00	95 min	Hands-on: Lab 13: Managing Processes and Scheduling with Cron; Lab 14: Package Management and Web Server Deployment (Slides 102–107)
13:00–13:30	30 min	Lunch break
13:30–15:10	100 min	Hands-on: Lab 15: Network Configuration and Troubleshooting (Slides 108–111)
15:10–17:15	125 min	Topic 4 — Linux Security: security principles, least privilege, firewall (concepts). Hands-on: Lab 16: Hardening Linux Security; Lab 17: Building a Security Audit Script (Slides 112–127)
17:15–17:30	15 min	Course summary, recommended next courses, course feedback and Q&A (Slides 128–134)

9:30am – 5:30pm · 450 minutes (7.5 instructional hours), with a 30-minute lunch.

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Lab Reference

Topic	Learning Outcome	Labs
Topic 01: Linux Fundamental	LO1	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Linux Shell Script	LO2	Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12
Topic 03: Linux Networking Configuration	LO3	Lab 13, Lab 14, Lab 15
Topic 04: Linux Security	LO4	Lab 16, Lab 17

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